INCH-POUND
MIL-PRF-18950A(OS)
26 May 1999
SUPERSEDING
MIL-PRF-18950(NOrd)
23 January 1956

# PERFORMANCE SPECIFICATION

# PROJECTILES, HIGH-CAPACITY, ANTI-AIRCRAFT, AND ILLUMINATING

*This specification is approved for use within the Naval Sea System Command Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification establishes the requirements for high-capacity, anti-aircraft and illuminating projectiles inert parts, and is augmented by drawings and specifications shown on the list of drawings and specifications particular to the item of the contract.

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

**2.2 Government documents.**

**2.2.1 Specifications.** The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1320
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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-18907 Bands, Projectile Rotating

MIL-P-18948 Projectiles, Painting, Marking, and Greasing of

MIL-PRF-18949 Projectiles, Illuminating, Localized Heat Treatment of

(Unless otherwise indicated, copies of DoD specifications are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

**2.2.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

2838457 Ultrasonic Test (UT) For 5 Inch Steel Projectile Bodies

(Application for copies of drawings should be addressed to: Commander, Naval Surface Warfare Center, Dahlgren Division, Attn: Code G52, 17320 Dahlgren Road, Dahlgren, VA 22448-5100.)

**2.3 Non-Government publications**. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 8 Metallic Materials, Tension Testing of (DoD adopted)

ASTM E 1742 Radiographic Examination (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)



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ANSI/ASQC Z1.4 Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(DoD activities may obtain copies of ANSI/ASQC Z1.4 from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. The private sector and other Government agencies may purchase copies from the American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202-4606.)

**2.4 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

**3.2 Material.** The material for the body and base plug shall be high grade homogeneous steel produced and fabricated by a forging process. The basic configuration is obtained by the steel being mechanically worked while above its critical temperature, with final machining where necessary. Sufficient top and bottom discard shall be taken from the ingot to ensure that the blooms, billets, or bars from which the projectiles are made are free from piping, segregation, flake, rolling, or forging tears or similar defects. When only one projectile body is made from a single ingot, the base shall be toward the bottom of the ingot.

**3.2.1 Mechanical properties.** Mechanical properties of the projectiles shall be as listed on the applicable drawings or as required in the appropriate specifications.

**3.2.2 Chemical composition.** Chemical composition of the projectile assembly components shall be as specified on the applicable drawings. Prior to manufacture of projectiles, the contractor shall submit a statement of the composition of the material proposed for use in the manufacture of the bodies, plugs, and bands to NAVSURFWARCEN, Crane for approval. This statement shall indicate the maximum and minimum percentages of alloying elements intentionally added or otherwise present in the steel. The chemical composition of all projectiles furnished under the contract shall be in accordance with this statement as approved by the contracting activity. Unless otherwise specified, the phosphorous and sulfur content shall each be limited to 0.05 percent maximum as determined by ladle analysis.

**3.2.3 Heat treatment.** All components shall be given uniform heat treatment to produce material conforming to the requirements of this specification. Components produced by band welding, cold forming operations or subjected to any cold work after heat treatment shall be adequately stress relieved to assure freedom from susceptibility to cracking or subsequent distortion as a result of unrelieved stresses. Hollow forgings shall not be quenched in water from above the critical



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temperature. The requirements of MIL-PRF-18949 shall govern the localized heat treatment required on illuminating projectiles.

**3.3 Surface finish.**

**3.3.1 Exterior.** All exterior surfaces shall have a smooth finish and shall be free from burrs, nicks, sharp edges, sharp corners, rough spots, or areas which fail to clean up. Bourrelets and other surfaces which make contact with the gun shall have a superior finish. Surface roughness shall not exceed the values specified on the applicable drawing.

**3.3.2 Cavities.** All cavities shall have smooth surfaces, free from burrs, nicks, sharp edges, sharp corners, rough spots or areas which fail to clean up. Cavities with surfaces as forged or cold formed are acceptable provided all requirements of finish, concentricities, dimensions, etc, are met.

**3.4 Shape.** The interior and exterior shape of the projectile shall be as shown on the applicable drawing.

**3.5 Interchangeability.** Components such as fuze hole plugs, adapters, and so forth which are not permanently assembled by the contractor or which, because of the requirements of fit or other considerations, are not clearly intended to be used in selective assembly, shall be entirely interchangeable.

**3.6 Band and band seat.** The projectile rotating band shall conform to MIL-PRF-18907. The band seat shall be machined before banding as shown on the applicable drawing.

**3.7 Marking.**

**3.7.1 Process marking.** The contractor shall mark projectile bodies and other components as required by the contract (see 6.2) to identify the component with its melt, ingot number, heat treatment, or other pertinent processes or stages of manufacture.

**3.7.2 Identification marking.** Finished projectiles shall be marked in accordance with the applicable drawing or MIL-P-18948 as specified in the contract (see 6.2). For projectiles assembled from bodies, base plugs or other components which are not entirely interchangeable, each component assembly shall be marked with the same serial number/zone symbol.

**3.8 Cleaning and preservation.** All surfaces of finished projectiles shall be cleaned and coated in accordance with the requirements of MIL-P-18948.

**3.9 Base plugs.** For projectiles with base plugs, the plugs and their mating counter bores in the projectiles shall be such that a gas-tight fit is obtained when the plug is assembled with a gasket. Unless otherwise specified on the drawings, interference fits between plugs and projectiles designed to have such a fit shall be $0.004 \pm 0.001$ inches. Base fuze hole plugs shall be made in such a manner that the longitudinal axis of the plug is normal (90°) to the fiber or grain flow lines of the material of



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which the plug is manufactured. When Acme threads are specified for a projectile and its mating base plug, special care shall be exercised to ensure the interchangeability of the parts. All threaded plugs shall be hand tight when shipped to the loading plant with the projectile and shall be readily removable upon arrival.

**3.10 Gaskets.** When gaskets are specified to be supplied under the contract (see 6.2), they shall be in accordance with the applicable drawing and specification. Unless otherwise specified, gaskets shall not be assembled with the projectiles but shall accompany each lot of projectiles; the number supplied shall be 10 percent in excess of the number of projectiles.

**3.11 Workmanship.** The workmanship shall be as specified on the applicable drawings, specifications, and the best industrial practices governing the quality production of interchangeable parts for maintaining the dimensions, finishes, tolerances, and quality specified herein. All components and assemblies shall be free from burrs, contamination, corrosion, sharp edges, or foreign material which could result in malfunction of the projectiles or be a safety hazard in handling.

## 4. VERIFICATION

**4.1 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

**4.2 First article inspection.** A first article sample of projectiles shall be inspected for conformance to the weight and dimension requirements of the applicable drawings. These projectiles shall be supplied in a bare metal condition with only a light coating of preservative oil or grease to prevent corrosion. A specified number of those projectiles shall also be tested in accordance with MIL-PRF-18907 and the applicable drawing, for conformance to 3.2.1. Additional first article projectiles shall be supplied in the phosphated and primed condition and subjected to the dry film thickness, adhesion, and salt spray tests of MIL-P-18948. The number of first article samples to be furnished in the different conditions and subjected to the various first article tests shall be as specified in the contract or order (see 6.2). Failure of any projectile shall cause rejection of the first article sample. Production of projectiles prior to acceptance of the first article inspection shall be at the contractor's risk.

**4.3 Quality conformance inspection.** The tests of 4.4 are required on all production lots. Any lot sampling shall be conducted in accordance with ANSI/ASQC Z1.4. For the first melt, a chemical analysis of the steel shall be forwarded to NAVSURFWARCEN Crane. For each additional melt, a chemical analysis shall be presented to the Government quality assurance representative. The analysis may be performed by the contractor or may be a certified analysis by the steel producer. The heat treatment of each lot, and the mechanical properties test results shall be recorded along with the result of dimensional inspections.



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**4.3.1 Production lot.** A production lot is a group of projectiles that are processed and heat treated as a group. No more than two mill heats of steel are allowed in one production lot and one mill heat of steel shall not be used in more than two production lots. The number of projectiles in the production lot shall be as specified in table 1.

TABLE I. Lot size.

| Caliber | Lot Size | |
|---|---|---|
| | Minimum | Maximum |
| 4" and smaller | 500/lot | 5000/lot |
| 5" | 500/lot | 2,500/lot |

**4.3.2 Inspection lot.** A grouping of one or more production lots submitted for inspection at the same time. All projectiles in an inspection lot shall be the same mill heat of steel. More than one heat treat may be combined in an inspection lot providing the heat treat process has been consistent over all these combined lots.

**4.4 Inspection and tests methods.** Instructions for inspection, tests, and manufacture of the projectile body are specified in the drawings and specifications and in 4.4.1 through 4.4.7.

**4.4.1 Samples for chemical analysis.** When specified in the contract (see 6.2), a chemical analysis of any bar or forging of any melt shall be made to determine the conformity of the material to 3.2.2.

**4.4.2 Fracture and macro-etch tests.** Sufficient discard from each ingot to ensure freedom from piping and segregation shall be determined by either a fracture test or a macro-etch test which shall be conducted on the cold stock representing the uppermost portion of each ingot. The fracture of macro-etched section shall show at least 10% of the cross-sectional area of the ingot, bloom, billet or bar; shall be at least 2" in diameter; shall be centrally located to show the axis of the ingot and shall be free from indications of piping or segregation. Discard shall be continued until the fracture or macro-etched section discloses satisfactory material. A fracture may be ground flat and suitably etched to indicate the condition of the material. Failure of the specimens to meet the requirements specified in 3.2 shall be cause for rejection of the lot.

**4.4.3 Tensile test.** The tensile test shall be performed on the projectile side wall as specified in ASTM E 8. Two, 5" and larger or four, 4" and smaller projectiles shall be randomly selected from each lot for tensile tests. Two or three test specimens per projectile, (or two per heat treat zone if differential heat treatment is required), as specified in the applicable projectile documentation, shall be cut at evenly spaced intervals around the circumference. Projectile lots may be re-heat treated and resubmitted for mechanical properties testing a maximum of two times. Six projectiles, two from the



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first third, two from the middle, and two from the last third of each heat treat group, shall be randomly selected from any lots resubmitted for testing. Failure of the test specimens to meet the tensile requirements specified in the applicable projectile documentation after reheat treatment shall be cause for rejection of the lot.

**4.4.4 Base plugs.** A sufficient number of base plugs shall be made from the same mill heat of steel and heat treated together so that all the plugs matched to a production lot of projectiles will be from the same production lot of plugs. Two base plugs shall be selected per production lot of plugs. A minimum of one test specimen shall be prepared from each plug. Base plugs may be re-heat treated and resubmitted for mechanical properties testing a maximum of two times. Six plugs shall be randomly selected from any lot resubmitted for testing. Failure of the specimens to meet the requirements specified after reheat treatment shall be cause for rejection of the lot.

**4.4.5 Ultrasonic testing.** All projectiles shall be subjected to ultrasonic testing in accordance with Drawing 2838457. The projectiles shall be examined for any structural defects. Any failures shall be rejected.

**4.4.6 Radiographic inspection.** Each projectile shall be radiographically inspected in accordance with ASTM E 1742, quality level 2-2T, in 2 directions rotated 90° apart and perpendicular to the longitudinal axis. Radiographs shall be examined for any structural defects. Any failures shall be rejected.

**4.5 Inspection of packaging.** The packaging, packing, and marking shall be visually examined for conformance to section 5.

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

**6.1 Intended use.** These military projectiles are intended for use with high explosive and illumination loads. They have no commercial application.



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**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).

c. Whether a first article sample is required (see 3.1).

d. Marking required (see 3.7.1 and 3.7.2).

e. If gaskets are to be supplied (see 3.10).

f. The number and condition of samples to be supplied for each first article test (see 4.2).

g. If chemical analysis of any bar or forging is required (see 4.4.1).

h. Packaging requirements (see 5.1).

**6.3 Subject term (key word) listing.**

Chemical composition
Metallic materials
Mechanical properties
Projectile rotating bands

**6.4 Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy (OS)
(Project 1320-0022)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-PRF-18950A(OS) | 2. DOCUMENT DATE (YYMMDD)<br>990526 |
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3. DOCUMENT TITLE
PROJECTILES, HIGH-CAPACITY, ANTI-AIRCRAFT, AND ILLUMINATING

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (*Last, First, Middle Initial*) | b. ORGANIZATION | |
|---|---|---|
| C. ADDRESS (*Include Zip Code*) | d. TELEPHONE (*Include Area Code*)<br>(1) Commercial<br><br>(2) DSN<br>(*If applicable*) | 7. DATE SUBMITTED<br>(*YYMMDD*) |

8. PREPARING ACTIVITY

| a. NAME<br>Commander, Indian Head Division<br>Naval Surface Warfare Center | b. TELEPHONE (*Include Area Code*)<br>(1) Commercial 301-744-4700<br>(2) DSN 354-4700 |
|---|---|
| c. ADDRESS (*Include Zip Code*)<br>Standardization Team (Code 840M)<br>101 Strauss Avenue<br>Indian Head, MD 20640-5035 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, VA 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 |

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